

7) N-Queens Problem by backtracking

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
int board[10], n;
```

```
int isSafe (int r, int c) {
```

```
    int i;
```

```
    for (i=1; i<=r; i++) {
```

```
        if (board[i] == c ||
```

```
            abs (board[i] - c) == abs (i - r)) {
```

```
            return 0;
```

```
        }
```

```
    }
```

```
    return 1;
```

```
}
```

```
void solve (int row) {
```

```
    int col, i;
```

```
    if (row > n) {
```

```
        for (i=1; i<=n; i++) {
```

```
            printf ("%d", board[i]);
```

```
printf("\n");  
return;  
}  
  
for (col = 1; col <= n; col++) {  
    if (isSafe(row, col)) {  
        board[row] = col;  
        solve(row + 1);  
    }  
}  
}
```

```
int main() {  
    printf("Enter no. of queens: ");  
    scanf("%d", &n);  
  
    solve(1);  
  
    return 0;  
}
```

O/p : Enter no. of queens: 4
2 4 1 3
3 1 4 2